

# curriculum vitae

Computational Mechanics Engineer with a Ph.D. in Computational & Experimental Mechanics, specializing in finite element analysis (FEA), multi-scale modeling, and experimental-numerical frameworks for microelectronics and advanced materials. Proven ability to develop simulation-driven solutions for characterizing mechanical behavior, identifying failure mechanisms, and supporting design optimization and reliability improvement. Strong background in semiconductor applications, thin-film mechanics, interfacial adhesion, and custom instrumentation.

## Personalia

|                 |                   |
|-----------------|-------------------|
| Surname:        | Doe               |
| Given names     | John              |
| Degrees:        | Ph.D.             |
| City:           | Jobville          |
| Phone number:   | 123 456 7890      |
| E-mail:         | johndoe@hmail.com |
| Date of birth:  | 1 January 1111    |
| Place of Birth: | Jobtown, Mexico   |
| Nationality:    | Mexican           |

## Professional experience

|                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2023 - Present | <b>Engineer, Laboratory</b><br><i>Lab</i> <ul style="list-style-type: none"><li>❖ Developed surrogate constitutive models and multi-scale simulation workflows to bridge microscale material behavior to engineering-scale predictions for nuclear fuel performance analysis.</li><li>❖ Built and validated advanced material models in finite element simulation tools.</li><li>❖ Created Python-based automation tools for large-scale simulation campaigns, data curation, and post-processing — significantly accelerating analysis turnaround time and improving reproducibility.</li></ul> |
| 2021 – 2023    | <b>Postdoctoral Researcher</b><br><i>Lab</i> <ul style="list-style-type: none"><li>❖ Built novel surrogate models for history-dependent material behavior (viscoplasticity and creep) based on regression methods applied to spectral (FFT) polycrystal simulation data.</li></ul>                                                                                                                                                                                                                                                                                                               |
| 2019 – 2020    | <b>Design Engineer</b><br><i>Lab</i> <p>Developed and maintained complex numerical algorithms and custom MATLAB tooling for high-precision nanoscale defect detection and geometry analysis on DUV lithography reticles. Designed, validated, and optimized metrology and data processing pipelines through large-scale data analysis and cleanroom experimentation, delivering production-ready solutions under tight deadlines in cross-functional industrial environments.</p>                                                                                                                |
| 2013 – 2019    | <b>PhD Researcher</b><br><i>University</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

## **Education**

University

|             |                                                                                      |
|-------------|--------------------------------------------------------------------------------------|
| 2013 – 2019 | <b>PhD. in Mechanical Engineering</b><br><i>Title: “test”</i>                        |
| 2013 – 2019 | <b>Graduate School of Engineering Mechanics</b><br><i>Advanced Mechanics Courses</i> |
| 2010 – 2013 | <b>MSc. Mechanical Engineering</b><br><i>Minor: Biomedical Engineering</i>           |
| 2006 – 2010 | <b>BSc. Mechanical Engineering</b>                                                   |
| 2000 – 2006 | <b>Research-oriented secondary education (VWO)</b>                                   |

## **Technical Skills**

|                                         |                                                                                                                                                                                                                                                                             |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Programming &amp; Software Tools</b> | Python, MATLAB, C++, Linux/bash scripting, SLURM (HPC), MSC Marc/Mentat, Paraview, LLM-assisted development (Continue.dev in VS Code)                                                                                                                                       |
| <b>Scientific Computing</b>             | Physics-based surrogate modeling, reduced-order modeling, spectral methods (FFT), finite element method (FEM), sparse linear solvers and regression, large-scale dataset curation & ablation studies, inverse optimization, uncertainty quantification, feature engineering |
| <b>Software Development</b>             | Custom CAE tool & pipeline development, workflow automation, interactive Python tooling (GUI), Python-to-C++ integration, production-grade code development, modular design, Git                                                                                            |
| <b>Domain Expertise</b>                 | Computational mechanics, multi-scale & multi-physics modeling, continuum mechanics, viscoplasticity and creep modeling, cohesive zone and interface mechanics, plasticity and dislocation mechanics                                                                         |
| <b>Technical Approaches</b>             | In-situ small-scale mechanical testing, high-resolution microscopy (SEM, SPM, optical profilometry), Digital Image Correlation (DIC), experimental-numerical frameworks                                                                                                     |

## **Selected journal publications (peer-reviewed)**